Ricardo Costa

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ABOUT

Software engineering student focused on building reliable, scalable software. Main interests include **software verification**, **developer tools**, and **agentic AI workflows**. Currently working on LiquidJava, a static analysis tool that detects violations of properties specified through Java annotations at compile time, such as valid value ranges and API protocols. Also experienced in backend systems, web and mobile applications, and cloud computing.

EXPERIENCE

LASIGE Research Unit 2025 – 2026 (Expected)
Software Engineering Researcher Lisbon, Portugal

Project: Improving the usability of LiquidJava, a liquid type system for Java, through enhanced verification feedback, extended IDE support, and a better overall developer experience.

* **Verification Feedback:** Redesigned diagnostics and implemented traceable predicate simplification, counterexamples, and actionable hints to make verification failures easier to understand and resolve.
* **IDE Integration:** Enhanced the VS Code extension with syntax highlighting, autocomplete, hover information, and interactive views for exploring diagnostics, inspecting the verification context, and visualizing typestate protocols.
* **Verifier Improvements:** Improved verification robustness and predictability through multiple error reporting, complete project verification, and closer alignment with Java syntax and semantics.

Contributions: 120+ merged pull requests to the verifier and 80+ to its VS Code extension.
Funding: FCT | LASIGE

EDUCATION

Faculdade de Ciências da Universidade de Lisboa (FCUL) 2024 – 2026 (Expected)
MSc in Computer Science and Engineering (Specialization in Software Engineering) Lisbon, Portugal

Current Grade: 17/20
Thesis: "Improving the Usability of LiquidJava"
Main Courses: Software Design (17/20), Compilation Techniques (18/20), Software Verification and Validation (16/20), Cloud Computing (18/20), Software Security (17/20), Mobile Computing (19/20)

Instituto Superior de Engenharia de Lisboa (ISEL) 2021 – 2024
BSc in Computer Science and Engineering Lisbon, Portugal

Final Grade: 15/20
Final Project: "NoteSpace: Multi-Platform Web Application for Real-Time Document Collaboration & Sharing" (18/20)
Main Courses: Programming (19/20), Software Laboratory (20/20), Algorithms and Data Structures (16/20), Web Programming (17/20), Mobile Programming (16/20)

PROJECTS

liquidjava | Improved verification workflow and feedback with better error messages and developer experience

vscode-liquidjava | Extended IDE support for reading, writing and debugging refinements in VS Code

liquidjava-mcp | Java MCP server that exposes LiquidJava verification tools to LLM agents over stdio

NoteSpace | Real-time collaborative editor with markdown and version control support

AI Code Reviewer | Lightweight AI code review workflow powered by Cloudflare Workers AI and KV

Barista | LLM agent workflow for synthesizing LiquidJava specifications from Javadoc documentation

Threadit | REST API with microservices architecture, gRPC service communication, and GCP deployment

aguda-rs | Compiler for a custom programming language implemented in Rust

passman | Secure multi-user CLI password manager with local encrypted storage built in Rust

Sketch.io | Multiplayer drawing and guessing game using React, Node.js, and Socket.IO

AdventOfCode | Solutions for Advent of Code coding puzzles using Python (88 completed)

SKILLS

Programming Languages | Python, Java, JavaScript, TypeScript, Kotlin, Rust, Go, C, C#
Backend | REST, gRPC, Microservices, WebSockets, Spring Boot, Node.js, Express, Socket.IO, Maven, Gradle
Cloud & DevOps | GCP, Cloudflare Workers, Docker, Kubernetes, Git, GitHub Actions, CI/CD, Linux
Frontend | React, HTML, CSS, Vite, Webpack, PWA
Mobile | Android, Jetpack Compose
Databases | PostgreSQL, Firestore, MongoDB, Cloudflare KV, Elasticsearch
Compilers & Static Analysis | Parsers, Type Systems, SMT Solvers, LLVM
Developer Tooling | VS Code Extension API, LSP, JSON-RPC
AI Engineering | AI Agents, Agentic Workflows, Cloudflare Workers AI, MCP

PUBLICATIONS

"Barista: Synthesizing Typestate Specifications with LLM Agents" | Catarina Gamboa, Paulo Canelas, Ricardo Costa, Márcio Caetano, Jonathan Aldrich, Alcides Fonseca (Accepted at Automated Software Engineering (ASE) 2026)

POSTERS

"Improving the Usability of LiquidJava" | Ricardo Costa, Catarina Gamboa, Alcides Fonseca
"Barista: Synthesizing Typestate Specifications with LLM Agents" | Catarina Gamboa, Paulo Canelas, Ricardo Costa, Márcio Caetano, Jonathan Aldrich, Alcides Fonseca (Best PhD Student Poster Award)

EVENTS

Poster Presenter | CIÊNCIAS Research & Innovation Day 2026 · "Improving the Usability of LiquidJava"
Speaker | JNation 2026 · "LiquidJava: Catch Bugs at Compile Time with Liquid Types"
Poster Presenter | 11th LASIGE Workshop · "Improving the Usability of LiquidJava"
Volunteer | Lisbon AI 2026